Topological Quantum Field Theory and the Jones Polynomial

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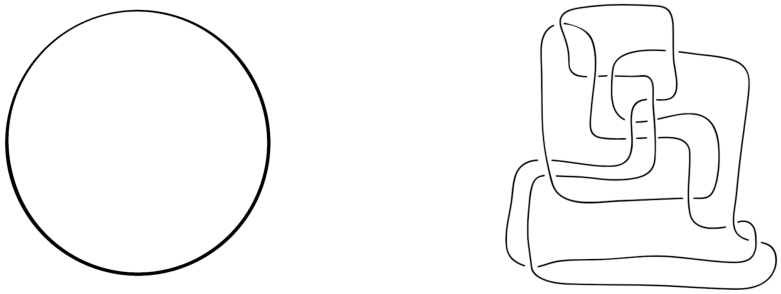
Knots & Links

A **knot** is an equivalence class of embeddings $K : S^1 \to S^3$ considered up to ambient isotopy.

A **link** is a disjoint union of knots considered up to ambient isotopy.

Topological Invariants

Given I have two knots or links, how does one determine whether they are distinct?



As in general topology, we distinguish knots/links by means of **topological invariants**.
This is where you'll list off the examples of the homotopy groups, homology, cohomology

Knot Invariants?

Ordinary (co)homology is useless for knots!

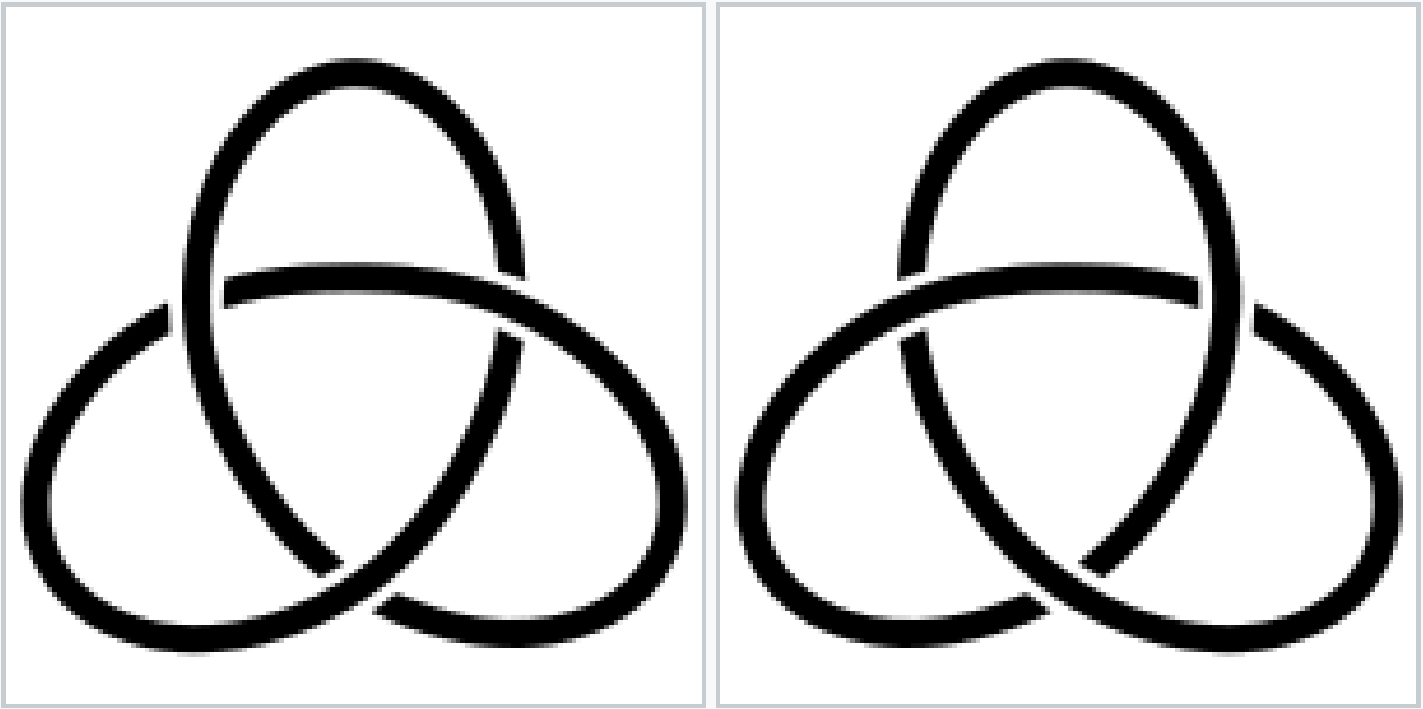
This is where you'll talk about how Alexander duality frustrates us.

$$\widetilde{H}_k(S^n - X) \cong \widetilde{H}^{n-k-1}(X).$$

The fundamental group of the complement of your knot/link is useful, but far from perfect.

Talk about how knots can be prime or composite. The fundamental group of the complement of a knot cannot tell which handedness the components are.

Fundamental group cannot detect knot parity



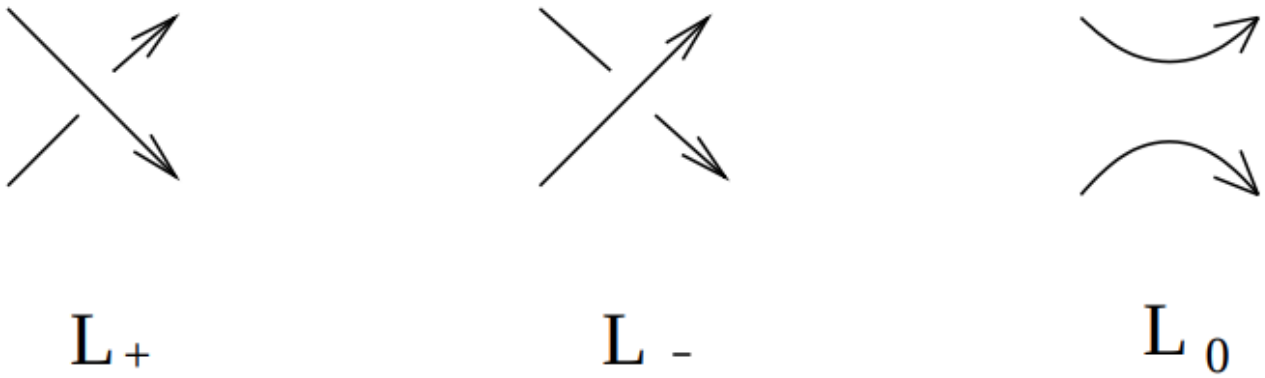
A left-handed trefoil and a right-handed trefoil.

Two trefoils of same parity gives Granny, opposite gives square.

Some mad science...

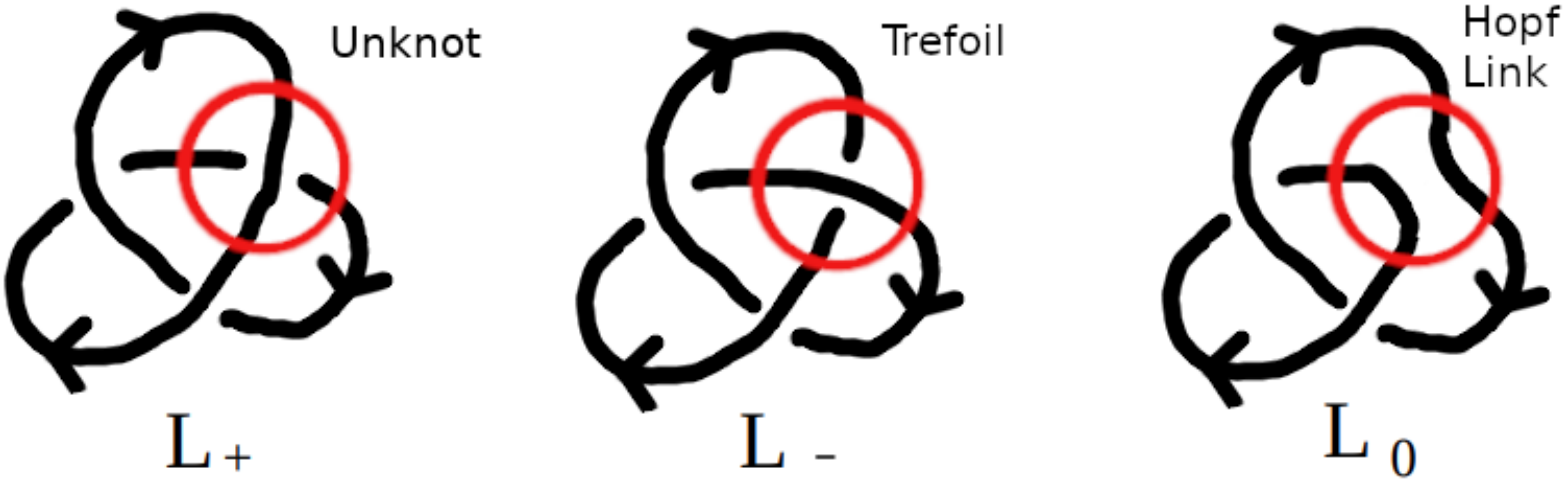
Given a link L, suppose we wanted to assign a polynomial J_L to it. How to get started?

Let's give L an orientation, choose a crossing, and play "mad scientist" by replacing it with one of the choices:



"Franken-links"

Picking the right-most crossing in the (left-handed) trefoil, we can create the following "Franken-links":



We'll use these "Franken-links" to define what's known as a **skein relation** that determines the knot polynomials.

The Jones polynomial

Letting $\mathfrak{L}$ denote the collection of all (tame) oriented links in $\mathbb{R}^3$, there is a unique map

$$J : \mathfrak{L} \to \mathbb{Z}[t, t^{-1}] \qquad J : L \mapsto J_L$$

satisfying

If L_1 and L_2 are isotopic, then $J_{L_1} = J_{L_2}$

If $P_L = 1$, then L is the unknot

The polynomials satisfy the skein relation:

$$t^{-1} P_{L_+} - t P_{L_-} = (t^{\frac{1}{2}} + t^{-\frac{1}{2}}) P_{L_0}$$

Some remarks

Why doesn't this depend on our choice of crossing?

There is an inductive argument for a generalized version of the Jones polynomial, called the HOMFLY polynomial.

Open problem

Is there a non-trivial knot with Jones polynomial equal to 1?

On this, problem Jones remarks:
"*One of the reasons that the question above has not been answered is presumably that, unlike... we have little intuitive understanding of the meaning ot the "t" in VL(t). Perhaps the most promising theory in this context is in where a complex is constructed whose Euler characteristic, in an appropriately graded sense, is the Jones polynomial. The homology of the complex is a finer invariant of links known as 'Khovanov homology'*"

Some examples

1. The disjoint union of two circles has Jones polynomial

$$t^{\frac{1}{2}} - t^{-\frac{1}{2}}$$

2. The Hopf link has Jones polynomial

$$t^{-1}P_{\text{Hopf Link}} - tP_{\text{Disjoint union of circles}} = (t^{\frac{1}{2}} - t^{-\frac{1}{2}})P_{\text{Unknot}}$$

$$P_{\text{Hopf}} = t\left(tP_{Circleunion} + (t^{\frac{1}{2}} + t^{-\frac{1}{2}})P_{\text{Unknot}}\right)$$

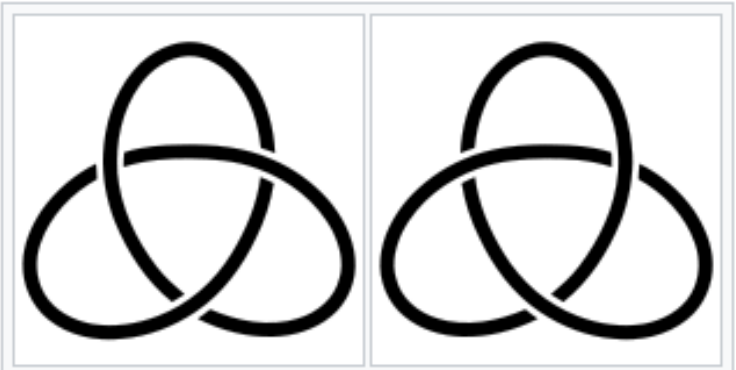
$$= t^2(t^{\frac{1}{2}} - t^{-\frac{1}{2}}) + t(t^{\frac{1}{2}} + t^{-\frac{1}{2}}) = t^{\frac{1}{2}}(1 + t^2)$$

3. Leveraging these, we compute the Jones polynomial of the left-handed trefoil to be

$$t^{-1} + t^{-3} - t^{-4}$$

The Jones polynomial detects knot / link parity

Observe that the Jones polynomial for the left and right handed trefoils are distinct! Moving between them, we replace t with t^{-1} and vice versa!



A left-handed trefoil and a right-handed trefoil.

Left-handed : $t^{-1}+t^{-3}-t^{-4}$

Right-handed : $t+t^{3}-t^{4}$

The cobordism category

We're going to make a category called **Cob(d)**:

1. Objects are closed, oriented d-manifolds.
2. A morphism $B : \Sigma_1 \to \Sigma_2$ is an equivalence class of $(d+1)$-manifold satisfying $\partial(B) = \overline{\Sigma_1} \sqcup \Sigma_2$ considered up to homeomorphism relative to the boundary.
   ($\overline{\Sigma_1}$ denotes Σ_1 but with reversed orientation)
3. We can compose morphisms by gluing
   Given $\partial B_1 = \overline{M} \sqcup N$ and $\partial B_2 = \overline{N} \sqcup O$,
   $B_2 \circ B_1 = B_1 \amalg_{N \sim \overline{N}} B_2$ and satisfies $\partial(B_2 \circ B_1) = \overline{M} \sqcup O$ and so is a morphism from M to O.
4. We regard the disjoint union of manifolds to be a product.

Topological Quantum Field Theory (Atiyah-Segel)

Fixing d and letting V_F denote the category of vector spaces over a field F, a **topological quantum field theory (TQFT)** is a functor

$$Z : \mathbf{Cob}(d) \to V_F$$

which satisfies

$$Z(\emptyset) = F \qquad Z(M \sqcup N) = Z(M) \otimes Z(N) \qquad Z(\overline{M}) = Z(M)^*$$

Remark on why this might represent something physical.

A few interesting remarks

We note that the axioms do not entirely determine the behavior of Z, but do impose strong conditions on possible behaviors.

Z sends the cobordism $\Sigma \times I$ to the linear map $i_\Sigma : Z(\Sigma) \to Z(\Sigma)$ since the boundary-gluing of two identical cylinders is homeomorphic to a single cylinder, it is clear that

$$i_\Sigma \circ i_\Sigma = i_\Sigma$$

If B is a closed $(d+1)$-manifold, then by definition $\partial B = \emptyset = \overline{\emptyset} \sqcup \emptyset$, meaning closed $(d+1)$-manifolds are sent to linear maps from the F to itself. Setting $F = \mathbb{C}$, this would assign to each closed $(d+1)$-manifold a complex number.

Witten's TQFT

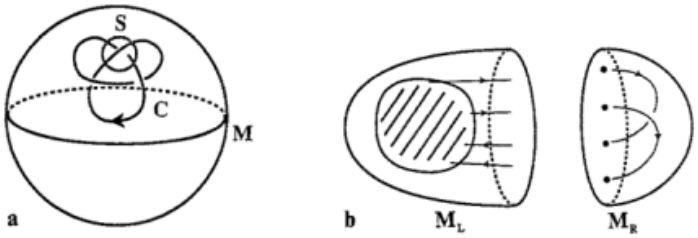
1. 3-manifolds $\rightarrow$ numbers / maps
2. 2-manifolds $\rightarrow$ vector spaces
3. 1-manifolds $\rightarrow$ vectors

Modify the theory so that instead of considering an assignment of a number to a closed $(d+1)$-manifold, we will instead perform such an assignment on a triple (B, L, μ_i) with B a 3-manifold, $L \subset B$ a link, and representation μ_i of a compact simple Lie group G for each component of L.

Think of the components of the link as representing the trajectories of particles and the Lie group representations as encoding some "state" (charge, spin, etc.)

Witten's TQFT

Given some link $L \subset S^3$, we can quarantine a crossing inside a 3-ball B^3. Denote the strands by $L' = L \cap B^3$. B and $S^3 - B^3$ meet at a copy of $\partial B^3 = S^2$ with marked points indicating the "in-going" and "out-going" of the strands in the link. Witten's 2-dimensional TQFT associates this S^2 with a vector space we'll call $\mathcal{H}$.



Then if μ_{i_1} and μ_{i_2} are representations associated with the strands comprising L', we should have $Z(B, L', \mu_{i_1}, \mu_{i_2})$ be some element $\chi \in \mathcal{H}$. (note that one dimensional vector subspaces are, up to scaling, just vectors.

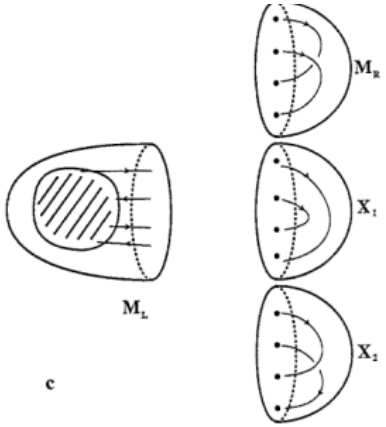
Meanwhile. $Z(S^3 - B, L - L', \mu_i)$ should be an element $\psi \in \mathcal{H}^*$. Functoriality tells us then that

$$Z(S^3, L, \mu_i) = \psi(\chi)$$

So iterate this over each of the crossing changes to get functionals α, β, γ. This means there is a linear dependence among outputs of the crossing changes!

$$(S^3, L_-, \mu_i) + \beta Z(S^3, L_+) + \gamma Z(S^3, L_0, \mu_i) = 0$$

This gives us an honest-to-god 3-dimensional description of the Jones polynomial!



Principle G-bundles and Connections

Given a manifold B and a topological group G, a G-**bundle** is a fiber bundle equipped with a regular action by G (i.e. to each point in the fiber, we can uniquely assign a group element).

Given a G a Lie group, we can generalize the notion of a connection on a vector bundle to create what's known as a **principle connection** on a principle G-bundle $B \xrightarrow{\pi} M$, which is a 1-form on B taking values in the Lie algebra $\mathfrak{g}$.

In a nut-shell, right multiplication by g induces a linear transformation on the Lie algebra and this introduces a relationship between objects in one fiber and objects in nearby fibers.

Holonomy of a connection

Letting $\pi : P \to M$ be a principle G-bundle and ω a principle connection, then given a smooth loop $\gamma : [0,1] \to M$ starting and ending at $m \in M$, the connection yields a unique lifting to the principle G-bundle $\widetilde{\gamma} : [0,1] \to P$ which starts a particular $p \in \pi^{-1}(m)$ and ends at some $p \cdot g \in \pi^{-1}(m)$.

We can then define

$$Hol_p(\omega) = \{g \in G \mid p \sim p \cdot g\}$$

Wilson loop

Given some $K \subset M$ be some closed curve,

$$W_r(C) = Tr_R P \; exp \int_C A_i dx^i$$

The invariants are given by the "vacuum expectation" vvalues of products of these Wilson loops:

$$Z(M, L, \mu_i) = \int D\mathcal{A} exp(i\mathcal{L}) \prod_{i=1}^{r} W_{R_i}(C_i)$$

Thank you!

Special thanks to the organizers and to my collaborator:

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